

DisplayGroundx
Capture Data Format & Dataset Verification Guide
User / Engineer Reference Documentation

1. Overview

DisplayGroundx processes datasets captured from a multi-band imaging sensor system. During acquisition, the capture system records:

- band image data
- acquisition parameters
- capture metadata

The captured data is stored as binary band files together with metadata files describing the acquisition. The application reconstructs frames from these files and allows visualization, analysis, and band selection.

2. Dataset Folder Structure

Example dataset:

Acq000315_07150928012026/

Acq000315_07150928012026.meta
Acq000315_07150928012026.log
Acq000315_07150928012026.json

Acq000315_07150928012026.band0
Acq000315_07150928012026.band1
Acq000315_07150928012026.band2
Acq000315_07150928012026.band3
Acq000315_07150928012026.band4
Acq000315_07150928012026.band5
Acq000315_07150928012026.band6

All files in a dataset share the same base name.

3. File Naming Convention

All capture files follow this format:

<base_name>.<extension>

Example base name:

Acq000315_07150928012026

File Types

.bandX -> Band image data
.log -> Acquisition log
.json -> Capture configuration
.meta -> Metadata information

4. Capture Parameter Line

Example:

001234 000070 001 11.02.2026 23:14:13.650 4.000000 127 0 14.000000 250.000000 1.000000 0 247 384

This line contains all parameters used to generate the dataset.

5. Parameter Breakdown

- 1 Orbit ID
- 2 Task ID
- 3 JSON ID
- 4 Date
- 5 Time
- 6 Duration (seconds)
- 7 Band Selection
- 8 TDI Setting
- 9 Frames per Second
- 10 Exposure
- 11 Gain
- 12 X Shift
- 13 Binning Configuration
- 14 TDI Y Shift

6. Orbit ID

Example: Orbit ID = 1234

7. Task ID

Example: Task ID = 70

8. JSON ID

Indicates the JSON configuration file used during capture.

9. Capture Time

Format:

DD.MM.YYYY HH:MM:SS.milliseconds

10. Capture Duration

Duration of capture in seconds.

11. Band Selection

Example:

127 = 01111111

Each bit represents a band.

Bit 0 -> Band0
Bit 1 -> Band1
Bit 2 -> Band2
Bit 3 -> Band3
Bit 4 -> Band4
Bit 5 -> Band5
Bit 6 -> Band6

01111111 means all 7 bands enabled.

12. TDI (Time Delay Integration)

Examples:

0 -> No TDI
10 -> TDI stage 2
18 -> TDI stage 4
34 -> TDI stage 8
66 -> TDI stage 64

13. Frame Rate

FPS = 14

14. Exposure

Exposure time applied by the sensor.

15. Gain

Gain = 1.0

16. X Shift

Horizontal pixel shift applied during capture.

17. Binning Configuration

Example:

247 = 11110111

Bit positions:

76543210

Bit 7 -> CCSDS compression flag

Bits 6-0 -> Band binning configuration

If a band is unbinned the system produces:

.bandX0 -> left half

.bandX1 -> right half

18. TDI Y Shift

Vertical shift applied during TDI processing.

19. Frame Count Calculation

frames = FPS × duration

Example:

FPS = 14

Duration = 4 seconds

frames = 14 × 4

frames = 56

20. Band File Size Calculation

Band File Size =
[(height / 2 if binned) / TDI stages] ×
(width / 2 if binned) ×
(bitdepth / 8) ×
number_of_frames

21. Bit Depth Storage

Example:
bit depth = 10 bits

$10 / 8 = 1.25$ bytes per pixel

22. Example File Size Calculation

width = 8448
height = 384
bit depth = 10
frames = 56

height_binned = $384 / 2$
width_binned = $8448 / 2$

23. Dataset Verification

Check:

- band files present
- metadata files available
- correct frame sizes
- consistent frame counts

24. Common Dataset Issues

- Missing band files
- Frame count mismatch
- Incorrect bit depth
- Black frames

25. Typical Processing Workflow

Capture Dataset
↓
Dataset Folder Created
↓
DisplayGroundx Opens Folder
↓
Application Detects Bands
↓
Frames Reconstructed
↓
Bands Visualized / Processed

26. Summary

A valid dataset requires:

- correctly named files
- consistent band files
- valid capture parameters
- correct bit depth
- matching frame counts